

What Kills? Factors that Affect Death Rate

Team 189



Introduction

National Health Expenditure (NHE) is one of the largest economic sectors in the United States accounting for 18.3% of the United States' Gross Domestic Product in 2021 (CMS, 2021) with NHE steadily growing since 1970 from 500 billion dollars to 4.3 trillion dollars in 2021 (Peterson-KKF Health System Tracker, 2021). National Health Expenditure is funded by federal taxes, taxes paid by the people of the United States, of which 29% is going to health programs and services spending (KFF, 2021). It is the goal of the United States Public Health Service to suggest the allocation of those funds in an effective manner that best serves the taxpayer. This important question serves as the basis for this study. This study highlights a few variables and the manner in which they contribute to the death rate in order to give insight into important problems and solutions that the United States Public Health Service should prioritize. The three research questions explored in this study are as follows: **Does higher Medicaid spending per capita lead to lower death rates? Does lower Medicare enrollment in states lead to higher death rates? Did lifting covid restrictions as of May 2021 (“yes”) result in higher death rates?** The rationale behind looking into the first two questions has to do with how the current national healthcare budget is built. As of 2021, Medicaid spending grew 9.2% to 734.0 billion dollars, or 17 percent of total NHE, and Medicare spending grew 8.4% to 900.8 billion dollars, or 21% of total NHE accounting for 38% of the total NHE budget (CMS, 2021). Due to this sizable amount of funding, it's important that the United States Public Health Service as well as the taxpayer is made aware of how effective these programs are (in terms of reducing death rates). And for the third question, understanding the effects of policy during the COVID-19 pandemic can help guide future solutions should a public health crisis arise again. Overall, the goal of this study is to provide recommendations for funding in the public health sector.

Methods and Analysis

The EDA portion of our analysis began with the quantitative variables; scatterplots and correlation coefficients were generated for each variable. The quantitative variables involved with two of our research questions (MedicareEnr and MedicaidSpend) showed little relationship with the response variable (DeathRate) in the scatterplots. Alternatively, the variable ObesityPrev showed a strong correlation with the response variable ($r=0.783$). After the EDA of the quantitative variables, we assessed those variables for multicollinearity. From this, we found that none of the quantitative variables had individual VIFs greater than 10 and the average VIF was less than 3, indicating that multicollinearity was not severe. Fol-

lowing this, we did stepwise regression to screen for variables with the pent value set at 0.10 and the prem value set at 0.30. We found ObesityPrev, OverdoseDeaths, and PopPerCap were added to the model while none were explicitly removed. Our higher order term (PopPerCap^2) was tested for significance through a t-test which proved to be significant with a p-value less than the alpha value of 0.05. With a final model for our quantitative variables, corresponding p-value for the Global F test was less than 0.05, indicating that our model was adequate. With regard to our qualitative variables, we created the following dummy variables; SeniorPopM (1 medium, 0 otherwise), SeniorPopH (1 high, 0 otherwise), PoliticalMajR (1 republican, 0 otherwise), and LiftCOVIDY (1 Yes, 0 otherwise). The EDA for these variables consisted of side-by-side boxplots. Since the senior population variable consisted of more than two levels, we performed a nested F-test, which yielded an insignificant p-value for the SeniorPop variable. We did not test for qualitative x quantitative interactions, as our interaction EDA did not suggest any relationships. With regard to our assumptions, the fit and normality assumptions were met while the constant variance assumption was violated with the individual residual plots for PopPerCap and PopPerCap^2 showing a clustering on the far left of the x-axis. In an attempt to correct this violation, we conducted transformations on the response variable in the form of square root, log, and the box-cox transformation, none of which were successful. This prompted a transformation of just the PopPerCap variable; this was a log transformation as is customary in this scenario. After the transformation, the final model consisted of the following equation: $\text{DeathRate} = \beta_0 + \beta_1 \text{ObesityPrev} + \beta_2 \text{OverdoseDeaths} + \beta_3 \log(\text{PopPerCap}) + \beta_4 \text{LiftCOVIDY}$. After this, we weighted each observation using the following formula: $w_i = \frac{1}{(\hat{\varepsilon}_i)^2}$. The rationale behind this was that statewide data comes with size differences that might affect the final model. With this weighted final model and log transformation, we met assumptions (See Appendix C for final model assumption plots). For the influential points and outliers, we found that Alaska and West Virginia were outliers in the x direction because their leverage surpassed our leverage cutoff of 0.16. We found that Delaware and Mississippi were outliers in the y-direction because their studentized residual values were outside of the ± 2 range. We also found that Delaware and Mississippi are influential points because their Cook's Distance values exceed our Cook's Distance cutoff of 0.08. We determined that we would not remove any of these points as we did not want to over-manipulate our results. Additionally, we found that the weighted least squares technique worked for our final model as the observations identified as outliers and influential points were a mix of different-sized states.

Results

In our comprehensive analysis of state death rates, we developed the following model: $DeathRate = \beta_0 + \beta_1 ObesityPrev + \beta_2 OverdoseDeaths + \beta_3 \log(PopPerCap) + \beta_4 LiftCOVIDY$. This final model consisted of statistically significant positive correlations between death rate and obesity prevalence, overdose death rate, and whether a state lifted COVID restrictions as of May 2021. It consisted of one statistically significant negative correlation between the logarithm of population per capita and death rate, evidenced by a negative beta estimate. The rest of the explanatory variables tested proved to be insignificant in helping predict the death rate in the model. This model accounts for 74.83% of the variation in the data with a RMSE value of 1.282 and a p-value of <0.00001 . Because the p-value of the model is less than 0.05, our model passes the Global F-test and therefore shows our model is adequate. It's also important to note that weighting the variables also increased the adjusted r-squared value and decreased the RMSE value from our unweighted final model.

Conclusions

Our final prediction equation is as follows: $\widehat{DeathRate} = 227.65 + 18.74 ObesityPrev + 4.20 OverdoseDeaths - 26.40 \log(PopPerCap) + 43.91 LiftCOVID$. As the prediction equation shows, for every one unit increase in obesity prevalence, the death rate per 100,000 residents increases by 18.74. For every one unit increase in drug overdose deaths per 100,000 residents, the death rate increases by 4.20. For every one unit increase in state population density, the death rate decreases by the logarithm of 26.40. The death rate for states with COVID restrictions lifted is 43.91 units higher than states that had not lifted restrictions. This model is useful for prediction. For instance, substituting values for the state of Virginia yielded a death rate of 954 people per 100,000 in 2021 while the actual value for the death rate in Virginia was 867 people per 100,000. This gives us a residual value of 87 and a percent error of 10%. This example shows us that our model is good for predicting death rates in different states during the year 2021. One limitation was that the transformation of the population per capita variable to a log sacrificed interpretability. For future research, we could select different policy-related variables such as air quality in urban areas and the median household income. Additionally, we could find other qualitative variables in order to not have to transform the senior population variable from a quantitative variable into a qualitative variable. Additionally, we would like to replicate this study using 2022 data to obtain the most recent data on what the death rate looks like.

Appendix A: Data Dictionary

Variable Name	Abbreviated Name	Description
Death Rate	Death.Rate	(Numeric) Death rate per 100,000 residents by state [<i>Response</i>]
Drug Overdose Death Rate	Drug.O verdose.Deaths	(Numeric) Drug overdose deaths per 100,000 residents by state
Obesity Prevalence	Obes ity.Prevalence	(Numeric) Prevalence of obesity based on self-reported weight and height by state
Total Medicare Enrollment	M edicare.Enroll	(Numeric) Medicare enrollment as a percent of the resident population by state
Total Medicaid Spending per Capita	Medicaid.Spend	(Numeric) Total Medicaid expenditure per capita (total divided by population) by state
Lifted COVID restrictions	Lift. COVID.Restrict	(Character/Categorical) If “yes”, then no general mask mandates, stay-at-home orders or interstate-quarantines in-place, most businesses open by state If “no”, restrictions still in place
State Population density	Pop.per.Cap	(Numeric) Population density in the U.S. by states

Variable Name	Abbreviated Name	Description
Senior Population Level	Senior.Pop	(Ch a racter/Categorical) Percentage of the state population with individuals 65 years or older split into three quantiles and labeled “Low”, “Medium”, and “High”.
Senior Population Level (Interval Form)	Senior.Pop1	(Factor) Senior Population Level shown in continuous interval form [11.6,16.6], (16.6,17.9], (17.9,21.7]. <i>*Not used in analysis, but included to support Senior.Pop</i>
Political Majority	Poli tical.Majority	(Ch a racter/Categorical) Assigns either “R” (for Republican) or “D” (for Democrat) based on voter data from the 2020 presidential election

Appendix B: Data Rows

	State	DeathRate	ObesityPrev	OverdoseDeaths	MedicareEnr	LiftCOVID
1	Alabama	1134.2	38.3	30.1	21.2	Yes
2	Alaska	931.0	32.1	35.6	14.8	Yes
3	Arizona	908.5	33.2	38.7	19.2	Yes
4	Arkansas	1097.8	37.4	22.3	21.6	Yes
5	California	760.4	28.1	26.6	16.6	No
6	Colorado	799.0	25.0	31.4	16.5	No

	SeniorPop	Senior.Pop1	PoliticalMaj	MedicaidSpend	PopPerCap
1	Medium	(16.6,17.9]	R	1320.066	99.52
2	Low	[11.6,16.6]	R	2957.729	1.28
3	High	(17.9,21.7]	R	2421.048	64.05
4	Medium	(16.6,17.9]	R	2368.123	58.16
5	Low	[11.6,16.6]	D	2798.871	251.90
6	Low	[11.6,16.6]	D	1856.213	56.08

Appendix C: Final Model Output and Plots

Call:

```
lm(formula = DeathRate ~ ObesityPrev + OverdoseDeaths + I(log(PopPerCap)) +  
    LiftCOVID, data = Data, weights = wt)
```

Weighted Residuals:

Min	1Q	Median	3Q	Max
-2.52336	-0.85707	-0.08865	0.92733	2.68912

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	217.2045	91.7413	2.368	0.02226	*
ObesityPrev	18.9840	2.7628	6.871	1.58e-08	***
OverdoseDeaths	4.1128	0.9137	4.501	4.74e-05	***
I(log(PopPerCap))	-25.3199	7.8361	-3.231	0.00231	**
LiftCOVIDYes	43.9053	25.5349	1.719	0.09241	.

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.27 on 45 degrees of freedom

Multiple R-squared: 0.7688, Adjusted R-squared: 0.7483

F-statistic: 37.41 on 4 and 45 DF, p-value: 8.933e-14

	StudRes	Hat	CookD
2	-1.1322038	0.21651647	0.07040902
6	2.0929452	0.24512261	0.26460347
11	-0.3957769	0.28292688	0.01259674
24	2.2795120	0.05669725	0.05713547
34	-2.1470286	0.07489623	0.06909783

Warning: 'aes_string()' was deprecated in ggplot2 3.0.0.

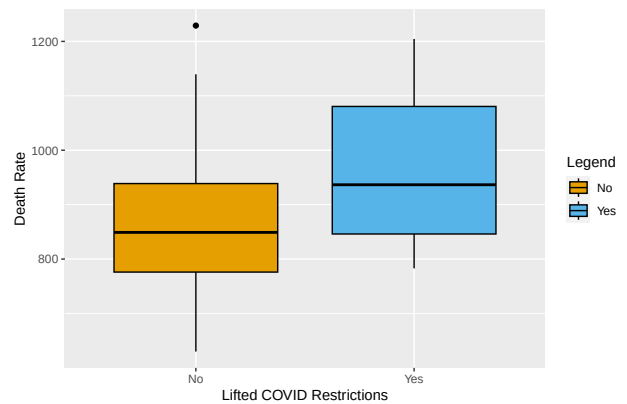
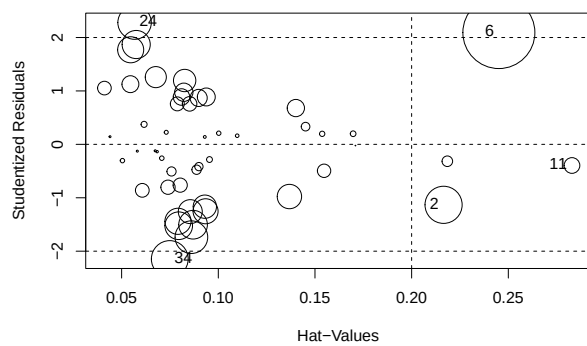
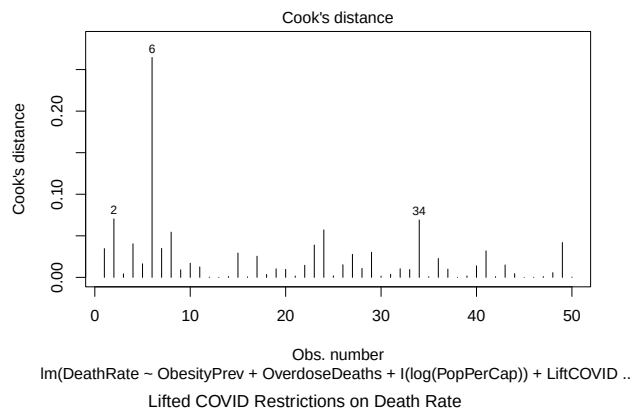
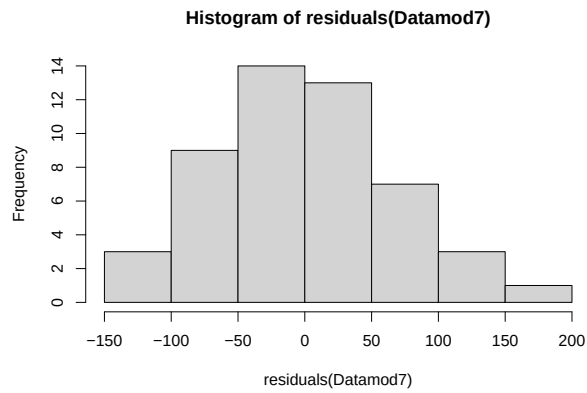
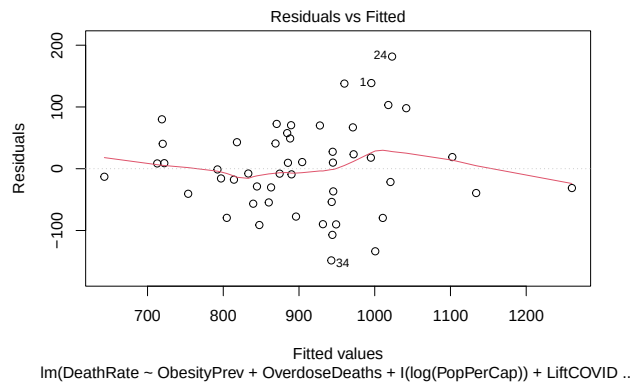
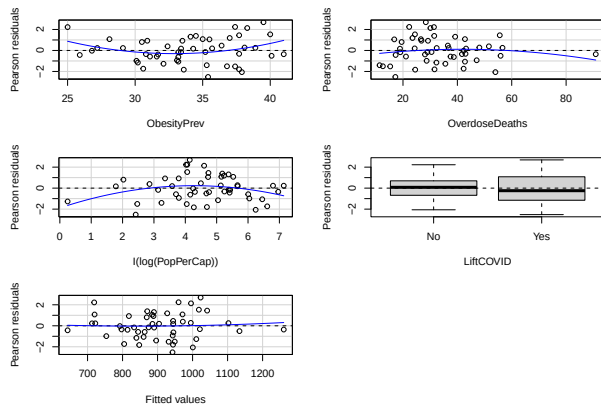
i Please use tidy evaluation idioms with 'aes()'.

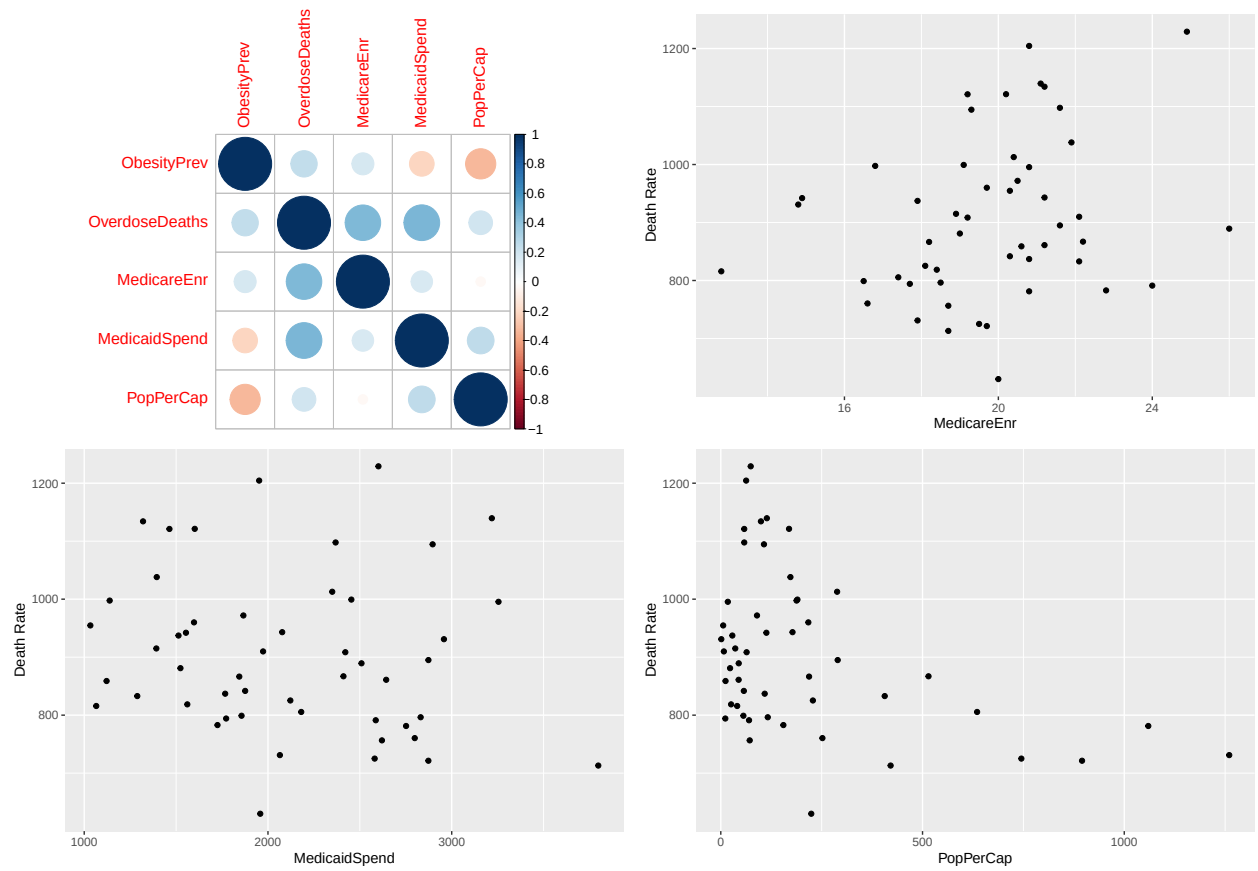
i See also 'vignette("ggplot2-in-packages")' for more information.

This warning is displayed once every 8 hours.

Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was

generated.





Appendix D: References

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11. Wasserman, D., Andrews, S., Saenger, L., Cohen, L., Flinn, A., & Tatarsky, G. (2020). 2020 National Popular Vote Tracker. Cook Political Report. <https://www.cookpolitical.com/2020-national-popular-vote-tracker>